

How does the new AI use so much energy?

Posted by u/extravagantsupernova • 0 points • 13 comments

What makes AI like ChatGPT use so much energy and resources compared to something like a regular search engine or wolframalpha

Comments

u/Real_Item_737 • 1 points • 2025-11-12 01:29 UTC

watch this! explains a LOTTT youtube.com/watch?v=-vwlQ_bvlW8&utm_source=substack&utm_medium=email

u/DiogenesKuon • 2 points • 2025-05-27 21:25 UTC

It uses a lot more energy than other similar computer operations traditionally do. One ChatGPT query is around 10x as much energy as a Google query (before google added AI to every response). But that's not actually a lot of energy in total. An average query costs somewhere around 0.4 watt-hours of energy. A modern LED lightbulb burns around 10 watt-hours per hour of use, so it's using around 2 and a half minutes of a single lightbulbs worth of energy. Forget to turn a couple of lights off when you head to bed is as wasteful as hundreds of AI queries.

So not really a big deal currently. But what happens if we keep shoving AI into everything imaginable, and the number of AI queries increases dramatically? Now we are starting to get into trouble, presuming we don't find more efficient ways to use AIs.

u/archpaw • 2 points • 2025-05-27 21:19 UTC

[It's not really that much.](https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about) They don't give official numbers, but it's estimated to be around 3 Wh. It's an order of magnitude more than a google search, but [lots of other things we don't really care about take the same] (https://docs.google.com/document/d/1pDdpPq3MyPdEAoTkho9YABZ0NBEhBH2v4EA98fm3pXQ/edit?tab=t.0#heading=h.k2ndl4fv47tw).

It also takes a huge amount to train the model, but that doesn't take as all the times it's used, so it's not a big deal either.

As for how it uses the amount it does, you need a big GPU to simulate all those neurons, and GPUs are power hogs.

u/VokThee • 1 points • 2025-05-27 21:16 UTC

Because what we call AI these days is just the same algorithms we've known for a long time, but a lot more of them running over a lot more data. There is nothing fundamentally more intelligent about AI

than before - it's just more and faster, to provide the illusion of intelligence.

u/TopNefariousness2306 • 1 points • 2025-05-27 21:16 UTC

AI uses tons of energy because it's basically a massive digital brain running millions of calculations non-stop—like your phone, but on caffeine and steroids.

u/Arbustri • 1 points • 2025-05-27 21:11 UTC

There's a nice analogy I heard from a professor on that. Imagine if regular search is similar to looking for the right book in a library, gen AI (ChatGPT) is closer to writing the book from scratch, so way more energy intensive.

u/GinosPizza • 3 points • 2025-05-27 21:10 UTC

LLMs like ChatGPT need a lot of compute power to run at large scale. This can only be supplied by large data centers. These are massive warehouses filled with servers (computers) that are all constantly running. That alone takes up a lot of electricity, but this also means there is a lot of heat. Most modern data centers also run elaborate cooling solutions to keep the servers cool. Those also suck up a lot of electricity.

u/itsatumbleweed • 1 points • 2025-05-27 21:57 UTC

I have a follow on comment. *A lot* of that energy goes into the training of the thing. A lot of the energy usage statistics take the training energy and aggregate it across the number of queries a model makes.

So a lot of the energy isn't necessarily because a single inference generates new heat, rather the training was expensive and keeping it running in a place that it can be queried is expensive.

u/Catalina_Eddie • 1 points • 2025-05-27 21:17 UTC



u/azuth89 • 1 points • 2025-05-27 21:09 UTC

Search engines just pull data from existing indexes, they don't generate anything new. AI digs through the data and generates a new summary of it.

It's the same way you can watch HD movies on a phone or a blue ray player with very little hardware because it's just showing things it already has, but you need a really powerful computer to live render graphics for a video game which, while amazing, still don't look as good as the stream/Blu-ray.

Because rendering something on demand needs more hardware, it takes up more power

u/Mairon12 • 1 points • 2025-05-27 21:09 UTC

Do you know what AI is?

It is for all intents and purposes the world's fastest and most accurate word predictor. That's putting it as basically as I can.

As the public ones do not yet use quantum computing, that type of efficiency requires a ton of energy.

u/itsatumbleweed • 1 points • 2025-05-27 21:54 UTC

Quantum computers also take a ton of energy- as of now they have to run at a temperature of approximately 0k (ELI5: 0 Kelvin is the coldest possible temperature. Coldness is really just an absence of heat, and 0k is no heat at all)... And they can't really do anything so impressive yet (don't get me started on "quantum supremacy" - there is a world where they are going to be revolutionary, but they need to start doing standard things better than regular computers to get that monicker).

QC at a not super cold setting isn't attainable yet.

u/Mairon12 • 1 points • 2025-05-27 21:55 UTC

Your information is quite outdated.

u/itsatumbleweed • 1 points • 2025-05-27 22:02 UTC

Feel free to correct me. Which bit? This was the case 3-4 years ago when "quantum supremacy" was popping up but I've not been working in that space for a while.

u/IgnitedIcedCoffee • 1 points • 2025-05-27 21:06 UTC

AI builds responses from scratch using complex math — way more energy-intensive than simple search or calculation tools.